When Is a Numerological Search Finished? An Exhaustive Null to Depth 10, and a Depth Ceiling That Explains Why Deeper Cannot Help

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Abstract

Searches for closed-form expressions reproducing Standard-Model dimensionless constants are common and almost never conclude: a null is reported as "we did not find one," which is not a statement about the space. We do two things about that, for one concrete search.

First, a worked exhaustive null. For a modified-inertia gravity framework whose acceleration scale is tied to the cosmological constant, `a0 = cH_Lambda/Z` with `Z = sqrt(32pi/3)`, we exhaustively enumerate every dimensionless value constructible at depth <= 10 from the framework's two forced germs — the generation count 3 and the kernel germ `sqrt(8pi/3)` — and confront all of them with 19 measured SM targets through three gates (germ content, cross-sector interlock, and false-discovery control). Result: **42,534,139 distinct values, 82,613 in-window hits, ZERO surviving the gate.** Depths 3–9 were already exhaustive clean nulls; depth 10 extends the proven-empty range and is the first depth past 9 where the word *exhaustive* applies rather than *sampled.*

Second, a depth ceiling. The diagnostic is not the null itself but *where the hits land*: hit count tracks measurement-window width and nothing else. The six most precisely measured targets return **exactly zero** hits across 42.5 million values; the loosely measured neutrino mixing angles return 15,000–26,000. Formalising that, a constructive enumeration whose candidate count grows as `B^(D-D0)` exhausts any *fixed* measurement precision `w` at a computable depth,

```
D_max  =  D0 + ln(1/w) / ln(B) ,
```

beyond which a single-target match is expected by chance and therefore carries no information however many digits it displays. For this vocabulary (`B = 30`, `D0 = 4`) the ceiling is **D = 10.4 for 1/alpha** and **D = 13.1 for m_p/m_e**, the most precisely measured ratio in physics. Depth 18 sampling — which we also ran — was statistically empty *by construction.* All discriminating power at depth must therefore come from *simultaneous* matching across independent sectors, with a threshold in bits: `SUM_i log2(1/w_i) > log2(N(D)) + margin`. Counting interlocked targets is the wrong statistic; three loosely measured targets carry 24.7 bits where two tight ones carry 76.3.

What is not claimed. The look-elsewhere/multiple-comparisons argument is standard (Gross & Vitells 2010); nothing here is new statistics. The contribution is its quantitative instantiation as a *depth* ceiling for constructive-enumeration searches, plus a worked exhaustive null. The null is about one germ vocabulary at one depth under one gate — it is not a statement about numerology in general, and not evidence for or against the framework whose germs it uses. We also report

an error of our own: the naive expected-count model `N*2w` **overestimates observed hits by ~100x** because the value set clusters rather than distributing uniformly, so our own bits thresholds should be read as conservative rather than exact.

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1. What was searched, and what "forced" means

The framework supplies exactly two germs and one free O(1) coefficient:

germ	origin in the framework
`3`	the generation count
`sqrt(8pi/3)`	the kernel germ, half of `Z = sqrt(32pi/3)`

"Forced" is load-bearing and is what makes the search falsifiable rather than open-ended. A candidate expression must *contain* the germs (Gate B); it is not enough to be some function of pi and small integers. Without that requirement the space is unbounded and no null means anything.

Expressions are built by constructive enumeration: a base leaf followed by `D-4` steps from a fixed 30-entry menu (leaf append with MUL/DIV, five power exponents, three unaries), decorated by canonical germ recipes. Candidate count therefore grows as `11 * 30^(D-4)`. Value-identical expressions are deduplicated at mpmath precision, so the object counted is *distinct values*, not distinct formulas.

Targets. 19 measured dimensionless SM quantities, spanning eleven orders of magnitude in relative precision — from `m_p/m_e` at 1.7e-11 to the PMNS mixing angles at ~6e-2. Two further targets (`koide_Q_lep`, `r_tau_mu`) were **held back** and never entered the search; §6.

2. The exhaustive null at depth <= 10

`sharded_build.py`, `sweep_depth.py`, `NULL_RESULT_DEPTH10_EXHAUSTIVE.md`.

```
raw candidates enumerated : 174,890,804
distinct values           :  42,534,139
targets swept             :          19   (held-back pair excluded)
in-window hits            :      82,613
CERTIFIED (gate-passing)  :           0
RE-LABELED                :           0
```

Depths 3–9 are exhaustive clean nulls under the same machinery. Depth 10 was previously attempted and abandoned partway; §7 describes what made it tractable.

3. The hit distribution is the result, not the null

target	rel. window	hits
`m_p/m_e`, `a_e`, `1/alpha`, `m_n/m_p`	1.7e-11 – 1.5e-10	**0**
`m_mu/m_e`, `a_mu`	2.2e-8 – 4.0e-7	**0**
`r_tau_e`, `1/alpha(M_Z)`, `sin^2 theta_W`	7e-5 – 1.7e-4	28 / 50 / 72
`koide_Q_up`, `higgs_lambda`, `ckm_lambda`, `koide_Q_down`	2e-3 – 5e-3	838 / 1130 / 2121 / 2098
`r_b_tau`, `r_t_b`, `alpha_s(M_Z)`	7.6e-3 – 1.1e-2	4747 / 4443 / 4933
`pmns_sin2_13/12/23`	3.5e-2 – 6e-2	15212 / 26142 / 20799

Hit count is a monotone function of window width and of nothing else. The six most precisely measured quantities in the set return **zero** hits across 42.5 million values; the least precisely measured return tens of thousands. Every one of the 82,613 died in the FDR gate. This is what chance looks like when it is drawn to scale, and it is a far more informative statement than "we found nothing."

4. The depth ceiling

`GATE_POWER_ANALYSIS.py`, `BITS_RULE.py`.

With `N(D)` candidates and a target of relative window `w`, the expected number of chance matches is `N(D)*w`; a hit is informative only when that is well below 1. Solving `N(D)*w = 1` for `D` with `N(D) = B^(D-D0)` gives the ceiling quoted in the abstract. For `B = 30`, `D0 = 4`:

target	rel. window	informative ceiling	expected chance hits at D=18
`1/alpha`	3.1e-10	**D = 10.4**	1.5e+11
`m_p/m_e`	3.5e-14	**D = 13.1**	1.7e+07
`alpha_s(M_Z)`	1.5e-2	D = 5.2	—

Two consequences.

(a) Deeper is worse, not better. Each additional depth costs `log2(B) = 4.9` bits of look-elsewhere for nothing, while available target precision is fixed. Depth 18 costs 68.7 bits; the two most precisely measured numbers in physics supply 76.3 between them, which does not clear it. Our own depth-10–18 sampling campaign (1,059 passes, ~29,000 in-window hits, zero survivors) was therefore incapable of establishing a single-target result *by construction* — a fact we did not appreciate until after running it.

(b) Interlock, weighted by bits. Requiring one expression to match `k` targets simultaneously multiplies the windows, so bits add. But *counting* targets is the wrong statistic: three loose targets give 24.7 bits, two tight ones 76.3. The threshold is `SUM_i log2(1/w_i) > log2(N(D)) + margin`, over targets a search is permitted to fit.

5. Why nothing was there — four independent structural obstructions

These were established separately and each would stand without the search:

1. **Number field.** `Z` carries a transcendental `sqrt(pi)`; the flavour and coupling data are algebraic. An exact identity therefore requires the `sqrt(pi)` to cancel, in which case the germ was not load-bearing.
2. **Period ring.** Half-integer versus integer weight, with the weight-1 slot empty — the two rings are disjoint exactly where a bridge would have to live.
3. **Dictionary/category.** `u = a0/g` is an acceleration *ratio*; a renormalisation scale is an *energy*. The only dimensional bridge, `a0/2c` with hbar and c, lands ~38 orders below the electron.
4. **Varying constants — the one experimental closure.** If any SM constant tracked the dark-energy density as strongly as `a0` does, atomic clocks would have seen it; the coupling is bounded to `|p| <= 6e-8`.

Four structural arguments plus the depth 3–9 nulls plus this depth-10 null plus a separate audit of a `12pi`-based alpha coincidence make seven independent lines reaching the same place.

6. Method: a held-back validation set, and why one of them is nearly worthless

`targets/pdg_constants.py` (`HOLDOUT_KEYS`).

A search that iterates until a gate passes is fitting to the gate. Two targets are therefore excluded from every search pool: `koide_Q_lep` (tight, rel ~1e-5) and `r_tau_mu` (loose, ~1e-4). A survivor would have to *predict* them.

Reported against interest: the tight one is nearly worthless as a holdout. Measured `koide_Q_lep = 0.666660511 +/- 6.8e-6` sits **0.91 sigma** from exact `2/3` — an answer known since Koide (1981). A survivor landing on `2/3` passes while predicting nothing new. `r_tau_mu` has no famous closed form and is therefore the stronger test despite being the looser one. Any holdout whose answer is already in the literature measures nothing.

7. Method: what made depth 10 tractable, and three bugs the guards caught

Depth 10′s cost is concentrated in a single budget split whose skeleton layer enumerates **8,019,000,000** step sequences at ~90,000/s — about 25 hours — and which collapses to only **22,708** distinct skeletons. Sharding the *search* alone does not help, because every shard rebuilds that same layer; the layer itself had to be parallelised and cached (`parallel_skeleton_layer.py`). With it cached, the depth-10 build runs in ~15 minutes.

Three bugs in the new code were caught by consistency guards rather than by inspection, and are recorded because the guards are the reason the null is trustworthy:

1. **A merge that re-keyed float64** silently discarded 19% of depth 8′s distinct values. Caught by requiring the sharded path to reproduce the committed serial counts *exactly* before being pointed at new depth.
2. **Shard-local record indices** made the hit records disagree with the merged value array. Caught by an existing `REBUILD MISMATCH` assertion in the committed sweep. Without it the null would have been computed against misaligned records.
3. **A holdout leak**: two hard-coded target lists re-added `koide_Q_lep` by name after the dataset-level exclusion, so the search was fitting a target documented as held back.

A fourth, in the opposite direction: applying the holdout filter unconditionally *broke the replay gate*, which must sweep the full historical target list to reproduce committed ground truth. A replay is a machinery check, not a search, and needs the holdout included. Suppressing a guard to satisfy a methodological rule is its own failure mode.

8. What this does and does not establish

Does. No expression built from these two forced germs at depth `<= 10` matches any of 19 SM targets with surplus information over chance, exhaustively. And: single-target matching in searches of this form is uninformative past a computable depth, so most such searches cannot conclude anything even in principle.

Does not. Say anything about depth 11+ (~13.5 days sharded) or 12 (~16 days) — and per §4 those depths could not carry a single-target result anyway. Say anything about other germ vocabularies. Bear on the framework whose germs were used: its acceleration scale, the constant `Z`, the response sign, and the gate frequency remain postulated, not derived, and this search neither supports nor damages them. Constitute new statistics: the look-elsewhere argument is standard, and our own instantiation of it was quantitatively wrong by ~100x until measured empirically (abstract).

Honest scale. This is one increment on a wall that already had six supports, obtained at substantial engineering cost, in a search whose prior was low and remains low. Its value is that it *closes* rather than *leaves ajar*: the range 3–10 is now proven empty rather than unexamined, and the ceiling says where examining further stops meaning anything.

References

Gross, E. & Vitells, O. 2010, *Eur. Phys. J. C* **70**, 525 — trial factors / the look-elsewhere effect. Koide, Y. 1981 — the charged-lepton mass relation `Q = 2/3`. Milgrom, M. 1999, *Phys. Lett. A* **253**, 273 — the kernel and identity the framework's germ derives from. Wilczek, F. — *Scaling Mount Planck*, Physics Today (dimensional-analysis numerology, context).